

ASSIGNMENT : 9

TITLE : Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement : Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives : To implement the final keyword with variables, methods, and classes.

Theory : The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

Exercise :

1. Create a **Bank Account program** where account number is final and cannot be changed.

```
class BankAccount {  
  
    final int accountNumber;  
    String accountHolder;  
    double balance;  
  
    BankAccount(int accountNumber, String accountHolder, double balance) {  
        this.accountNumber = accountNumber;  
        this.accountHolder = accountHolder;  
        this.balance = balance;  
    }  
  
    void display() {  
        System.out.println("Account Number: " + accountNumber);  
        System.out.println("Account Holder: " + accountHolder);  
        System.out.println("Balance: ₹" + balance);  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
  
        BankAccount acc = new BankAccount(10101, "Anushree", 25000);  
        acc.display();  
    }  
}
```

Output

Generated files

```
Account Number: 10101
Account Holder: Anushree
Balance: ₹25000.0
|
```

2. Create a **Library Book Management** program where the **book ISBN** is declared as **final** and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```
class Book {
    final String isbn;
    String title;
    String author;
    double price;

    Book(String isbn, String title, String author, double price) {
        this.isbn = isbn;
        this.title = title;
        this.author = author;
        this.price = price;
    }

    void display() {
        System.out.println("ISBN: " + isbn);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: ₹" + price);
    }
}

public class Main {
    public static void main(String[] args) {
        Book b = new Book("3842349", "Java Basics", "James Gosling", 650);
        b.display();
    }
}
```

Output

Generated files

```
ISBN: 3842349  
Title: Java Basics  
Author: James Gosling  
Price: ₹650.0
```

Outcome :

- Understood the use of the **final** keyword with variables, methods, and classes.
- Implemented immutable values using final variables.
- Learned how **final** improves program reliability and data security.

Conclusion :

This assignment demonstrated the use of the **final** keyword in Java. It prevents unwanted modifications to variables, methods, and classes, thereby improving data integrity, security, and application reliability.